

adds Deshpande.

"The school market has the potential to grow bigger than the current engineering-college market. We have a wide range of products intended for different demographic sections. We have introduced robotic kits for school kids (in the age group of 10-17 years) and have been promoting them passionately. We are getting an overwhelming response from parents as well as school authorities," says Deshpande.

Also, thanks to the progressive policies of the government, government-aided engineering institutes are open to introducing innovative lab products as well as conducting skill enhancement courses for their students.

The budget has provided lots of opportunities. The Indian government levies 3 per cent of education cess in central excise for spending on education, and there is a tremendous increase in the number of technical institutes all over India.

"The last decade saw an exponential growth of degree/diploma educational institutes in the technical field with the number of institutes increasing to 3500+ in 2011. This, coupled with technological changes, stringent accreditation rules, competition among themselves and government policies on quality improvement, has resulted in constant growth of the educational kits sector and we are sure that the trend will continue in years to come," says Joshi.

Looking into the future, present technologies along with prospective ones (especially embedded, automation, telecom and networking) need to be coupled with e-learning to provide educators with tools having perfect

"A well-equipped lab is a boon to the students. We wish the government to come up with schemes and regulations for the same—Mohammed Ghouse, country manager (business communication), Sciencetech Technologies

Robotics is no longer just a hobby. Students are becoming serious and want career-oriented training—Adwait Deshpande, R&D Head, Robotics, Thinklabs

price-performance-quality balance.

Are we equipped to tap them?

For large-scale adoption of educational kits, fund allocation is a major bottleneck. It is not that the funds are unavailable. The only requirement is that the management of private colleges, which constitute over 75 per cent of the total colleges, should wake up to the fact that to be in competition, quality improvement is necessary.

In most of the industries, a part of the profit is spent on development activities. Educational institutions, on the other hand, refrain from spending on new products. The staff may not recommend new products and even if it does, it fails to convince the management about the necessity of the new product and cost.

"Majority of the engineering colleges buy kits just to fulfil the requirements of the syllabus. There is least scope of R&D for the students, says B.M. Shah, chief executive officer, Ed-kits Electronics.

"Colleges starting BE/B.Tech degree courses in ECE, EEE and E&I streams need to invest in basic labs, but many colleges in Tier 3 category compromise even on basic equipment. Basically, the labs should provide general-purpose equipment like power supplies, digital storage oscilloscopes, digital multimeters, function generators, RF signal sources, spectrum analysers and vector network analysers," says V.V. Pathy, business development manager-education, Agilent Technologies, Bengaluru.

He adds, "Agilent provides such colleges with teaching labs at a subsidised price. These labs provide a set of equipment (for example, digital multimeters, power supplies, function generators and digital storage



Universal robotics development platform from Kits'n'Spares

oscilloscopes) along with a lab kit and courseware."

Agilent also has solutions for the postgraduate or research level, in varied fields like RF, microwave, wireless, digital, semiconductor, optical and nanotechnology.

What needs to be done

Though there is a strong potential for educational and hobby kits in India, this segment requires attention from the government as well as educational institutes to become one of the major contributors to national income.

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Tax relief to the industry would help in effective cost-control and therefore large-scale adoption of newer technologies. Similarly, rebates on import and export duties will act as a strong support for the manufacturers.

For players, "I would suggest that simply duplicating and selling simplicity will not take this industry to newer heights unless everyone develops some technically sound kits that can enhance the knowledge of users," says Sahu.

"Companies involved in the manufacture of educational and hobby kits will have to keep themselves abreast with the latest technological advancements in technical education. They also need to emphasise more on R&D to make the kits more interesting and educative," adds Agarwal. ●

The author is executive editor at EFY